

# Elimination with Matrices

How every computer solves  $Ax = b$ :  
eliminate down to a triangle, then **back-substitute**.

Shown on a real money question: splitting ₹10 lakh across funds.

Pivots & multipliers

Portfolio mix

MIT 18.06, Lecture 2

## THE PROBLEM

# Pictures don't scale

In Lesson 1 we solved a portfolio by drawing lines and spotting tricks. With 50 funds and 50 rules, nobody can draw or guess.

The same three-fund system, now solved properly:

$$\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 10 \\ 111 \\ 7.4 \end{bmatrix} \quad \begin{array}{l} \text{budget} \\ \text{return} \\ \text{exposure} \end{array}$$

## THE METHOD

**Elimination:** the way every software package solves equations.

# Triangles are easy

$$x + y + z = 10$$

$$-5y + 3z = -9 \quad \Rightarrow \quad z = 2, y = 3, x = 5$$

$$-0.4z = -0.8$$

Solve the last equation, then work upwards.  
Elimination turns any good system into a triangle like this, **without changing its solution**.

## TWO WORDS

**Pivot:** the diagonal entry used to clear the entries below it.

**Multiplier:** entry to clear  $\div$  pivot.

2 × 2

## Two funds, one step

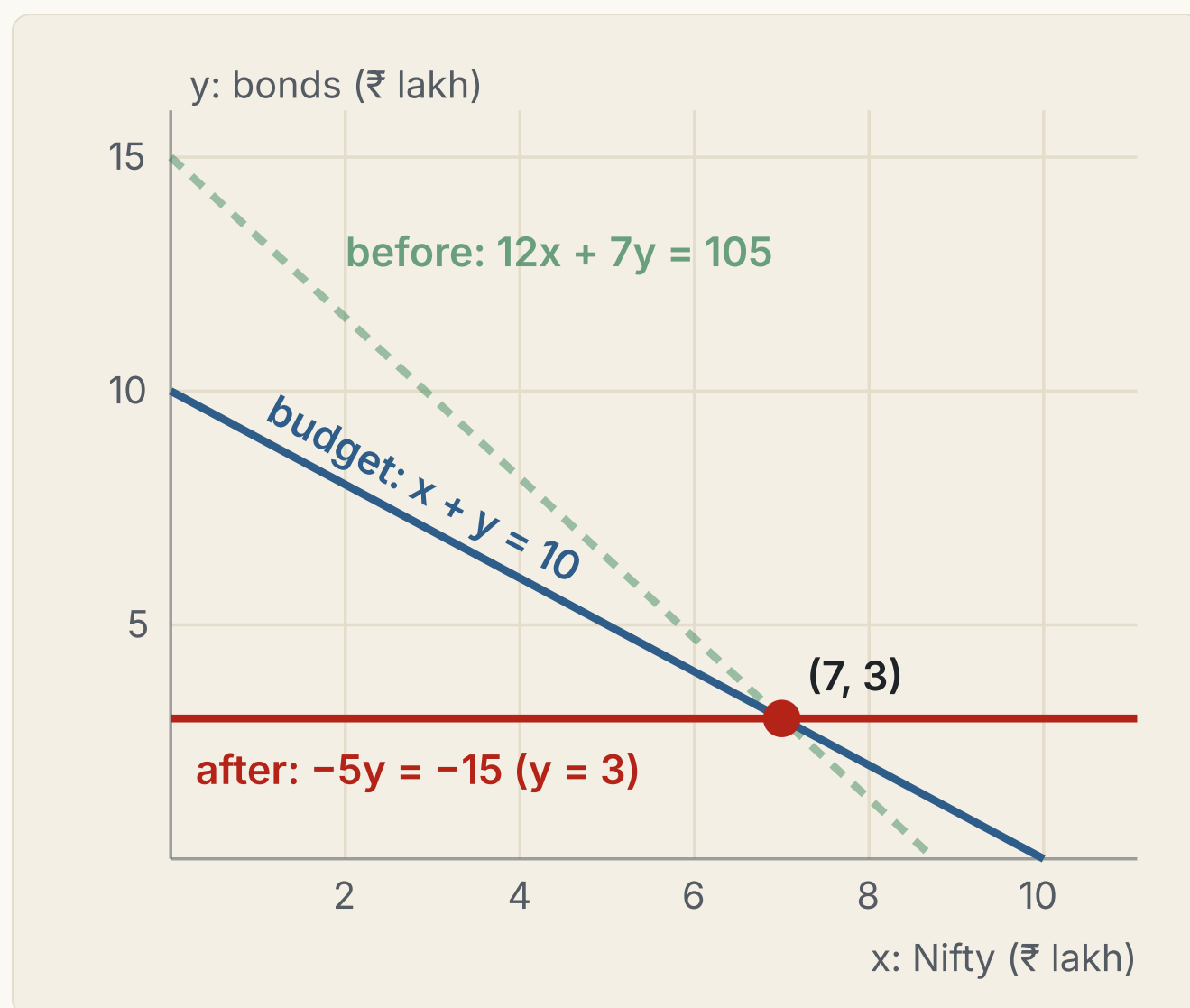
$$\left[ \begin{array}{cc|c} \boxed{1} & 1 & 10 \\ 12 & 7 & 105 \end{array} \right] \xrightarrow{R_2 - 12R_1} \left[ \begin{array}{cc|c} \boxed{1} & 1 & 10 \\ 0 & \boxed{-5} & -15 \end{array} \right]$$

Multiplier  $12/1 = 12$ . Subtract  $12 \times$  the budget rule from the return rule:  $-5y = -15$ .

So  $y = 3$ , then  $x = 7$ : **₹7 lakh Nifty, ₹3 lakh bonds.**

WHAT CHANGED?

# New line, same answer



Elimination swaps the return line for  $y = 3$ . The crossing point  $(7, 3)$  doesn't move.

# Clear below the first pivot

$$\left[ \begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 15 & 111 \\ 1 & 0 & 1.2 & 7.4 \end{array} \right]$$

- $\ell_{21} = 12$ : row 2 – 12 × row 1
- $\ell_{31} = 1$ : row 3 – row 1

$$\left[ \begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & -5 & 3 & -9 \\ 0 & -1 & 0.2 & -2.6 \end{array} \right]$$

# Clear below the second pivot

$\ell_{32} = (-1)/(-5) = 0.2$ : row 3  $- 0.2 \times$  row 2.

$$[U \mid c] = \left[ \begin{array}{ccc|c} \boxed{1} & 1 & 1 & 10 \\ 0 & \boxed{-5} & 3 & -9 \\ 0 & 0 & \boxed{-0.4} & -0.8 \end{array} \right]$$

## RESULT

$A$  became upper triangular  $U$ ;  $b$  became  $c$ . Same solution. Pivots 1,  $-5$ ,  $-0.4$  on the diagonal.

# Solve from the bottom up

$$1. -0.4z = -0.8 \Rightarrow z = 2$$

$$2. -5y + 3(2) = -9 \Rightarrow y = 3$$

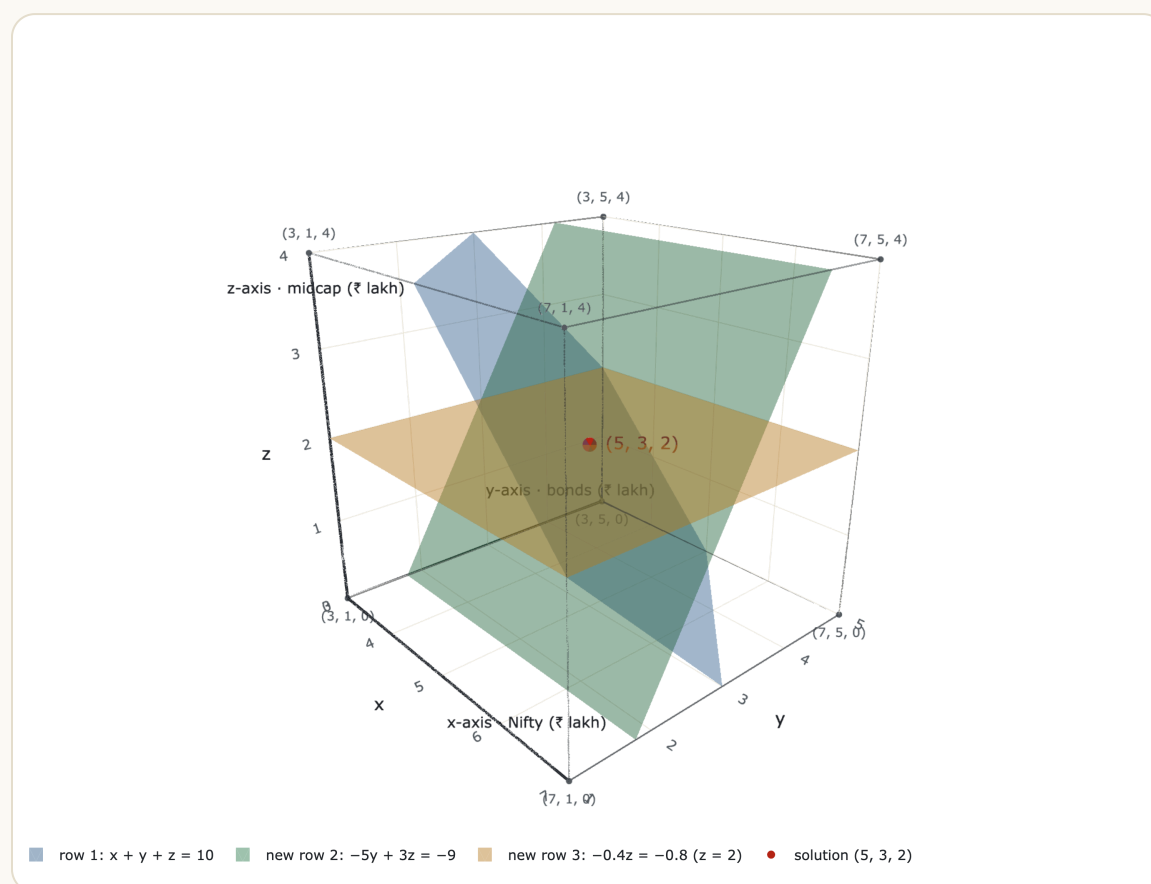
$$3. x + 3 + 2 = 10 \Rightarrow x = 5$$

₹5 lakh Nifty, ₹3 lakh bonds, ₹2 lakh midcap.

## THE WHOLE METHOD

Forward elimination, then back substitution. That's all a computer does, with millions of rows.

# After elimination



New planes, ending with the flat  $z = 2$ . They still meet at the same point  $(5, 3, 2)$ .

# A zero in the pivot position

**Temporary:** a non-zero entry below it? Exchange rows and carry on. (A credit fund earning 12% like Nifty causes this.)

**Permanent:** nothing non-zero below? No pivot: the matrix is **singular**. (A 50/50 balanced fund: the last row becomes  $0 = -0.8$ .)

## PIVOTS DECIDE EVERYTHING

$n \times n$  matrix:  $n$  pivots  $\Leftrightarrow$  invertible. Fewer: no solution or infinitely many.

# Each step is a multiplication

Row  $\times$  matrix combines rows:  $\begin{bmatrix} -12 & 1 & 0 \end{bmatrix} A = -12 (\text{row } 1) + (\text{row } 2).$

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -12 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{32}E_{31}E_{21}A = U$$

Parentheses can move (associative), so  $E = E_{32}E_{31}E_{21}$  does all of elimination at once.

**BUT ORDER MATTERS**

$AB \neq BA$  in general.

# Swaps and undo

**Permutation  $P$ :** the identity with rows swapped.  
 $PA$  swaps rows of  $A$ ;  $AP$  swaps columns.

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{adds back what } E_{21} \text{ subtracted}$$

## WHY IT WORKS

Every step can be undone, so  $Ax = b$  and  $Ux = c$  have exactly the same solutions.

# Key takeaways

1. Elimination turns  $Ax = b$  into triangular  $Ux = c$ , same solution.
2. Multiplier = entry  $\div$  pivot. Pivots can't be 0.
3. Back substitution: last row first.
4. Zero pivot: swap rows, or the matrix is singular.
5. Each step is a matrix:  $E_{32}E_{31}E_{21}A = U$ .
6. Steps are reversible, which is why it's right.

QUANT FROM FIRST PRINCIPLES

# The full interactive lesson is free

A lab that runs elimination step by step on your own numbers, 3D plots and worked examples:

**[learn.krisarj.com](https://learn.krisarj.com)**

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